EXERCISE1 : CAEAR CIPHER

AIM :-

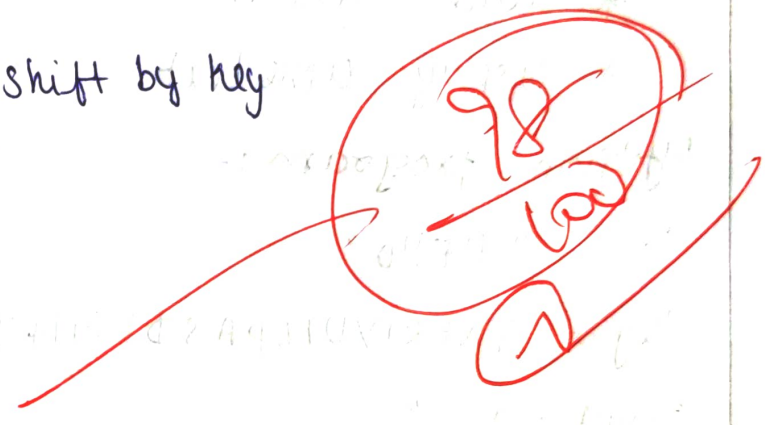
To write a program to implement the caesar cipher by shifting each letter by a given key.

Algorithm :-

* Read the plaintext and key value.
* Shift each alphabet character by the key.
* use modulo 26 to wrap around the alphabet.

pseudocode :

* Read text and key
* For each character, shift by key
* Display ciphertext.



program :-

```
text = "Hello"
key = 3
result = " "
for ch in text:
    result += chr(ord(ch) - 65 + key) % 26 + 65)
print("plaintext: ", text)
print("ciphertext: ", result)
```

Output :

Plaintext : HELlo

ciphertext : KHOOR

Result :-

Thus, the caesar cipher was implemented successfully.

Exercise 2: Monoalphabetic substitution cipher

AIM:

To write a python program to implement a monoalphabetic substitution cipher.

Algorithm:-

* Take the plaintext and substitution alphabet
* Display the ciphertext

pseudocode:

* Read plaintext and substitution key
* Display ciphertext

python program:-

```
text = "HELLO"
key = "QWERTYUIOPASDFGHJKLZXCVBNM"
result = ""
for ch in text:
    result += key[ord(ch)-65]
print("plaintext :", text)
print("ciphertext :", result)
```

Output:

Plaintext : HELLO
ciphertext : ITSSG

Result:

Thus, the monoalphabetic substitution cipher was implemented successfully.

Exercise 3 - playfair cipher

Aim :-

To write a python program to demonstrate the playfair cipher using a 5×5 matrix.

Algorithm :-

* create a 5×5 matrix using the keyword
* Applying the playfair encryption rules

pseudocode:

Divide plaintext into pairs
Apply playfair rules.

python program :-

```
matrix = [ "MONAR",
           "CHYBY",
           "EFGIK",
           "LPQST",
           "UVWXZ" ]
print("playfair matrix :")
for row in matrix :
    print(row)
print("plaintext : "HELLO")
print("playfair matrix created successfully")
```

output: playfair matrix : MONAR
CHYBY
EFGIK
LPQST
UVWXZ

plaintext : HELLO
playfair matrix : matrix created successfully.

Exercise : 4 polyalphabetic substitution cipher.

Aim:

To write a python program to implement a polyalphabetic

Algorithm :-

* Take plaintext and key
* Add plaintext and key values.

pseudocode :-

Repeat key over plaintext
Add corresponding values.

python program :-

```
text = "Hello"
key = "KEY"
result = ""
for i in range (len (text)):
    result += chr (ord (text[i]) - 65 + ord (key [i % 3]) - 65) % 26 + 65)
print ("plaintext :", text)
print ("ciphertext :", result)
```

output:

Plaintext : Hello
ciphertext : RIJVS

Result:

Thus, the polyalphabetic subsitution cipher was implemented successfully.

Exercise 5: AFFINE CIPHER

AIM:

To write a python to implement the affine cipher using the formula $c = (ap+b) \bmod 26$.

Algorithm:

* Take plaintext and values of a and b
* convert the result back into letters

pseudocode:

Calculate $c = (ap+b) \bmod 26$

Display ciphertext

Python program:

```
text = "Hello"
a,b = 5,8
result = ""
for ch in text:
  p = ord(ch) - 65
  result += chr((a * p+b) % 26 + 65)
print("plaintext :", text)
print("Ciphertext :", result)
```

Output:

plaintext : HELLO

ciphertext : RCLLA

Result:

Thus, the affine cipher was implemented succesfully.

Exercise: 6 - KEYWORD MONOALPHABETIC CIPHER

AIM:

To write a python program to implement a monoalphabetic cipher using a keyword.

Algorithm:

* match plaintext letters with the cipher alphabet
* Replace every plaintext letter

pseudocode:

* Create cipher alphabet
* Replace plaintext letters.

Python program:

```
text = "HELLO"
key = "CIPHERABDFGIJKLMNOQSTUVWXYZ"
result = ""
for ch in text:
    result += key[ord(ch) - 65]
print("plaintext :", text)
print("ciphertext :", result)
```

Output:

```
plaintext : HELLO
ciphertext : EATTA
```

Result:

Thus, the keyword-based monoalphabtic cipher was implemented successfully.

Exercise: 7 - playfair matrix encryption

AIM:

To write a python program to use the given playfair matrix for encryption.

Algorithm:

* Divide message into pairs
* Encrypt using matrix

Python program:

```
matrix = [
    "MFHIK",
    "UNOPQ",
    "ZVWXY",
    "ELARG",
    "DSTBC"
]
print("playfair matrix:")
for row in matrix
print (row)
print ("message : MUST SEE YOU")
```

output:

playfair matrix :

MFHIK
UNOPQ
ZVWXY
ELARG
DSTBC

message : MUST SEE YOU

Result:

Thus, the given playfair matrix was implemented successfully.